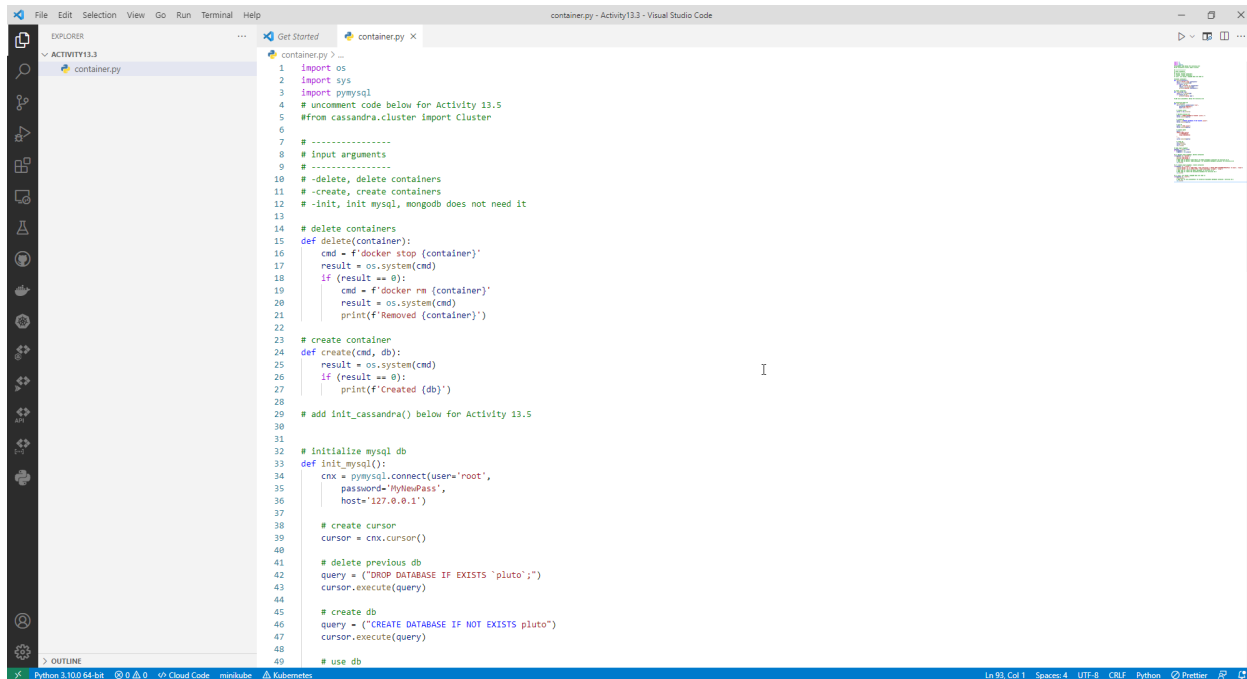


Activity 13.3 Solution File

1.

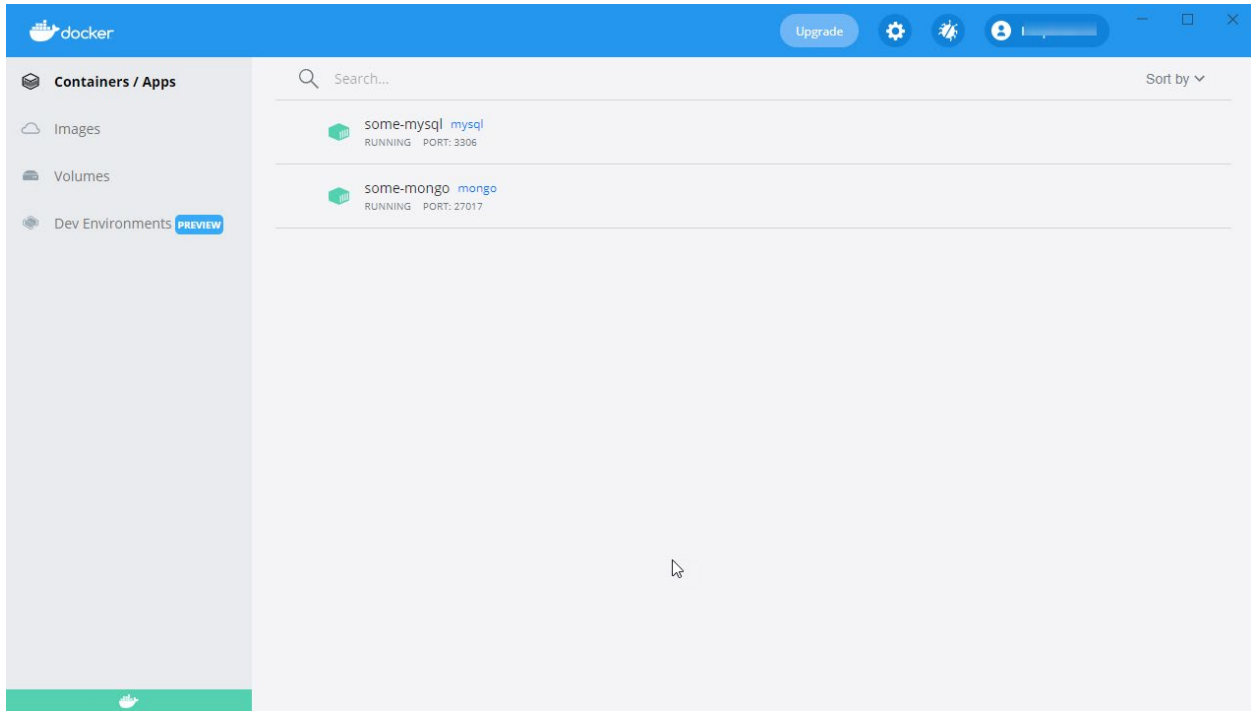


```
1 import os
2 import sys
3 import pymysql
4 # uncomment code below for Activity 13.5
5 #from cassandra.cluster import Cluster
6
7 # -----
8 # input arguments
9 # -----
10 # -delete, delete containers
11 # -create, create containers
12 # -init, init mysql, mongodb does not need it
13
14 # delete containers
15 def delete(container):
16     cmd = f'docker stop {container}'
17     result = os.system(cmd)
18     if (result == 0):
19         cmd = f'docker rm {container}'
20         result = os.system(cmd)
21         print(f'Removed {container}')
22
23 # create container
24 def create(cmd, db):
25     result = os.system(cmd)
26     if (result == 0):
27         print(f'Created {db}')
28
29 # add init_cassandra() below for Activity 13.5
30
31
32 # initialize mysql db
33 def init_mysql():
34     cnx = pymysql.connect(user='root',
35                           password='MyNewPass',
36                           host='127.0.0.1')
37
38     # create cursor
39     cursor = cnx.cursor()
40
41     # delete previous db
42     query = ("DROP DATABASE IF EXISTS 'pluto';")
43     cursor.execute(query)
44
45     # create db
46     query = ("CREATE DATABASE IF NOT EXISTS pluto")
47     cursor.execute(query)
48
49     # use db
```

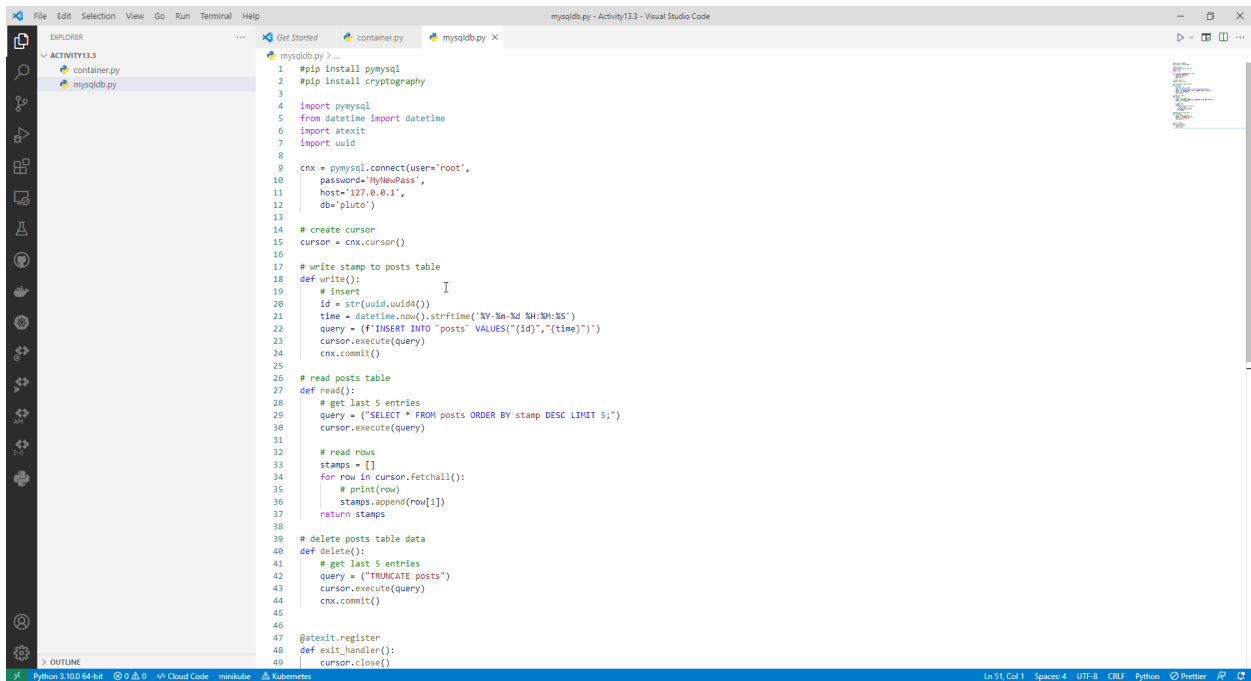
2.

```
Windows PowerShell
PS D:\MO-PCDE\Module13\Activity13.3> python container.py -create
Traceback (most recent call last):
  File "D:\MO-PCDE\Module13\Activity13.3\container.py", line 3, in <module>
    import mysql.connector
ModuleNotFoundError: No module named 'mysql'
PS D:\MO-PCDE\Module13\Activity13.3> python container.py -create
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
b380bbd43752: Pull complete
f23cbf2ecc5d: Pull complete
30cfc6c29c0a: Pull complete
b38609286cbe: Pull complete
8211d9e66cd6: Pull complete
2313f9eeca4a: Pull complete
7eb487d00da0: Pull complete
4d7421c8152e: Pull complete
77f3d8811a28: Pull complete
cce755338cba: Pull complete
69b753046b9f: Pull complete
b2e64b0ab53c: Pull complete
Digest: sha256:6d7d4524463fe6e2b893ffc2b89543c81dec7ef82fb2020a1b27606666464d87
Status: Downloaded newer image for mysql:latest
8eae6215beab9ab87b340345f6e5199cdf0b486f7a296da50fa78182a0bd870f
Created mysql
Unable to find image 'mongo:latest' locally
latest: Pulling from library/mongo
7b1a6ab2e44d: Pull complete
90eb44ebc60b: Pull complete
5085b59f2efb: Pull complete
c7499923d022: Pull complete
019496b6c44a: Pull complete
c0df4f407f69: Pull complete
351daa315b6c: Pull complete
557b07ecd9d7: Pull complete
a2dff157a5e3: Pull complete
07d83e88231b: Pull complete
Digest: sha256:07212fb304ea36b8c5a9e5694527f16deeb0b99f87fc60162dc15ab260bf8a2a
Status: Downloaded newer image for mongo:latest
e8896afdfaa71014dea7a9ae646fde8babd46903597c086d0cde496a27668f55
Created mongo
PS D:\MO-PCDE\Module13\Activity13.3> 
```

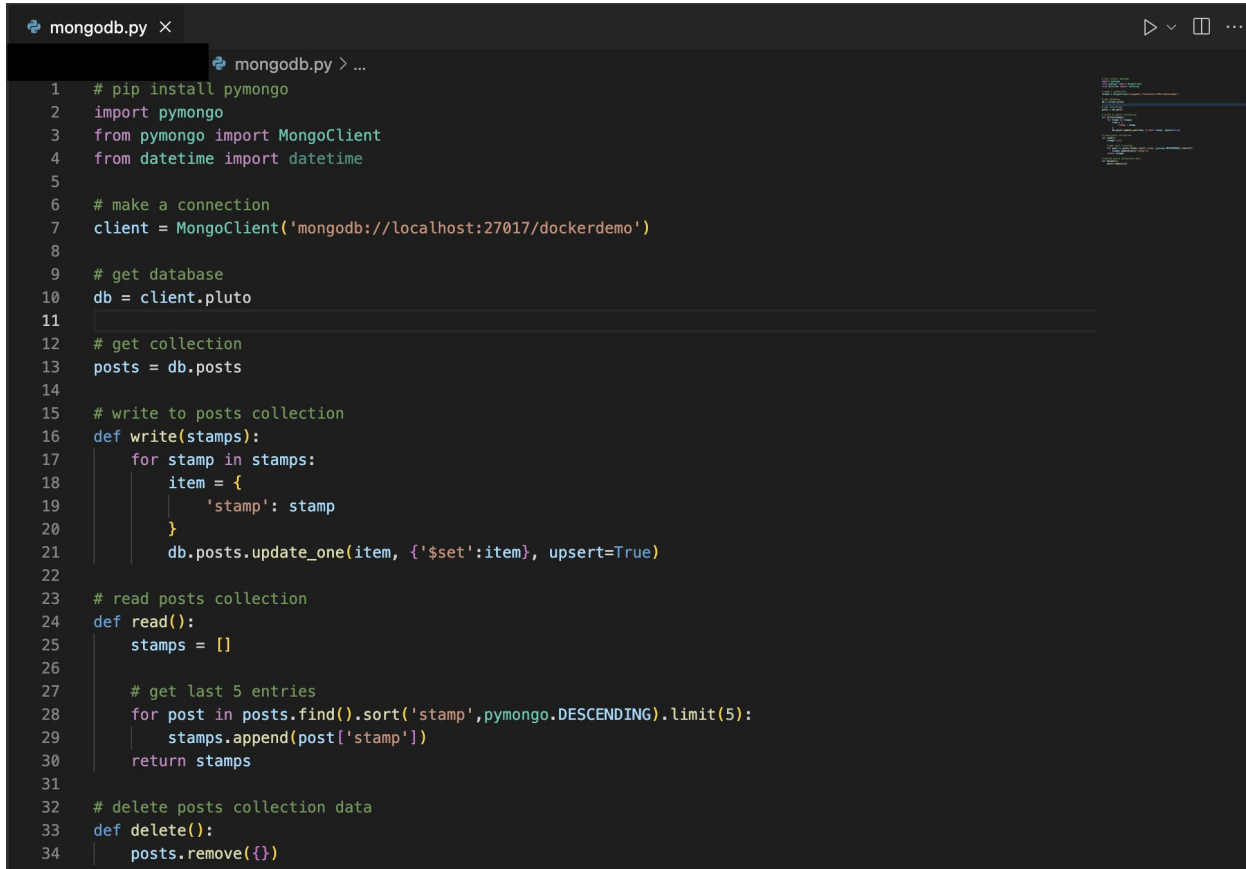
3.



4.

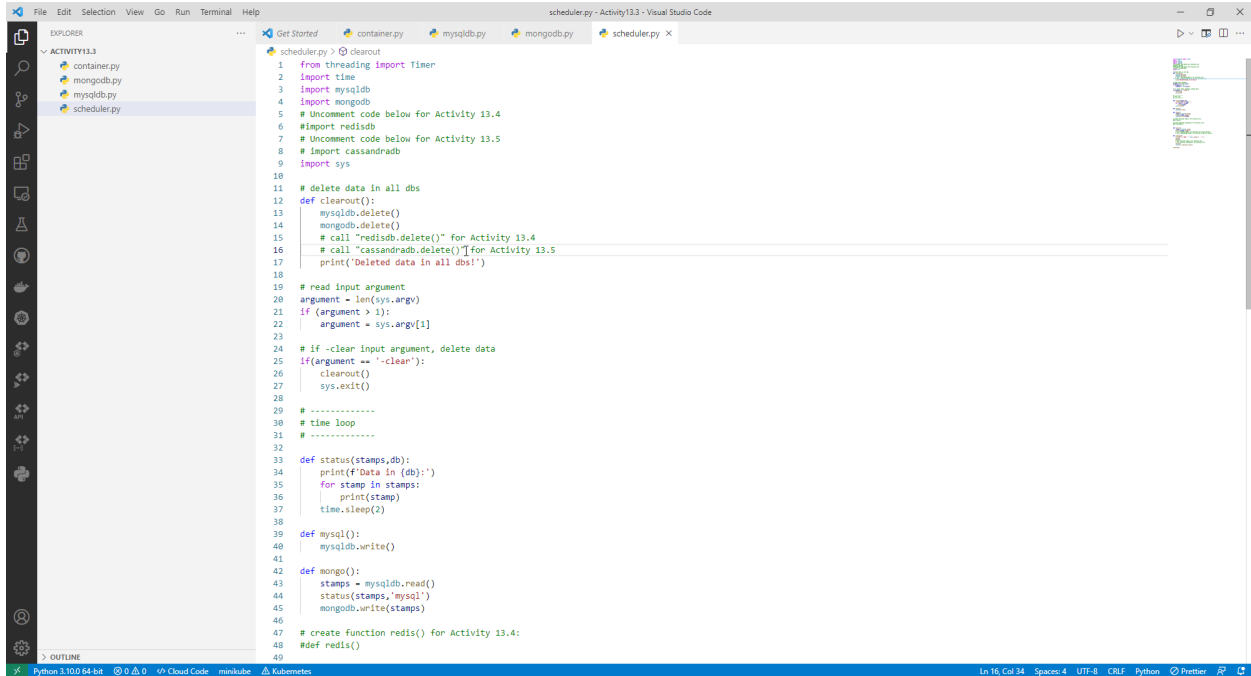


5.

A screenshot of a code editor window titled 'mongodb.py'. The editor shows a Python script for interacting with MongoDB using the pymongo library. The script includes comments and code for connecting to a MongoDB instance, creating a database and collection, writing data, reading data, and deleting data. The code is as follows:

```
1 # pip install pymongo
2 import pymongo
3 from pymongo import MongoClient
4 from datetime import datetime
5
6 # make a connection
7 client = MongoClient('mongodb://localhost:27017/dockerdemo')
8
9 # get database
10 db = client.pluto
11
12 # get collection
13 posts = db.posts
14
15 # write to posts collection
16 def write(stamps):
17     for stamp in stamps:
18         item = {
19             'stamp': stamp
20         }
21         db.posts.update_one(item, {'$set': item}, upsert=True)
22
23 # read posts collection
24 def read():
25     stamps = []
26
27     # get last 5 entries
28     for post in posts.find().sort('stamp', pymongo.DESCENDING).limit(5):
29         stamps.append(post['stamp'])
30     return stamps
31
32 # delete posts collection data
33 def delete():
34     posts.remove({})
```

6.

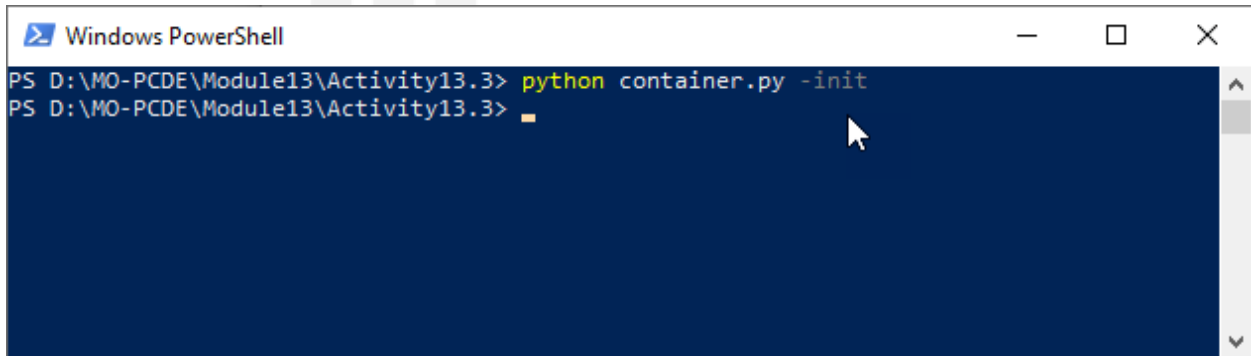


```

1 from threading import Timer
2 import time
3 import mysql
4 import mongo
5 # Uncomment code below for Activity 13.4
6 #import redis
7 # Uncomment code below for Activity 13.5
8 # import cassandrdb
9 import sys
10
11 # delete data in all dbs
12 def clearout():
13     mysql.delete()
14     mongo.delete()
15     # call "redis.delete()" for Activity 13.4
16     # call "cassandrdb.delete()" for Activity 13.5
17     print("Deleted data in all dbs!")
18
19 # read input argument
20 argument = len(sys.argv)
21 if (argument > 1):
22     argument = sys.argv[1]
23
24 # if -clear input argument, delete data
25 if(argument == "-clear"):
26     clearout()
27     sys.exit()
28
29 # -----
30 # time loop
31 # -----
32
33 def status(stamps,db):
34     print(f'Data in {db}:')
35     for stamp in stamps:
36         print(stamp)
37     time.sleep(2)
38
39 def mysql():
40     mysql.write()
41
42 def mongo():
43     stamps = mysql.read()
44     status(stamps,mysql)
45     mongo.write(stamps)
46
47 # create function redis() for Activity 13.4:
48 #def redis():
49

```

7.



```

PS D:\MO-PCDE\Module13\Activity13.3> python container.py -init
PS D:\MO-PCDE\Module13\Activity13.3>

```

8.

```

Windows PowerShell
no 2] No such file or directory
PS D:\MO-PCDE\Module13\Activity13.3> python scheduler.py
--- LOOP: Fri Nov 5 04:30:51 2021 ---
Data in mysql:
2021-11-05 04:30:51
Data in mongo:
2021-11-05 04:30:51
--- LOOP: Fri Nov 5 04:31:00 2021 ---
Data in mysql:
2021-11-05 04:31:00
2021-11-05 04:30:51
Data in mongo:
2021-11-05 04:31:00
2021-11-05 04:30:51
--- LOOP: Fri Nov 5 04:31:09 2021 ---
Data in mysql:
2021-11-05 04:31:09
2021-11-05 04:31:00
2021-11-05 04:30:51
Data in mongo:
2021-11-05 04:31:09
2021-11-05 04:31:00
2021-11-05 04:30:51
--- LOOP: Fri Nov 5 04:31:18 2021 ---
Data in mysql:
2021-11-05 04:31:18
2021-11-05 04:31:09
2021-11-05 04:31:00
2021-11-05 04:30:51
Data in mongo:
2021-11-05 04:31:18
2021-11-05 04:31:09
2021-11-05 04:31:00
2021-11-05 04:30:51

```

9.

```

Windows PowerShell
PS D:\MO-PCDE\Module13\Activity13.4> python container.py -delete
some-mysql
some-mysql
Removed some-mysql
some-mongo
some-mongo
Removed some-mongo
Error response from daemon: No such container: some-redis
PS D:\MO-PCDE\Module13\Activity13.4>

```

10.

